Vibe-IC — ALL Steps: Phase → Stage → Step (v2.2.0)

Every flow step, organized as **Phase → Stage → Step**, with one continuous main-track number **1 → 41** (starting at Stage 1, Spec-to-RTL). Phase 1's doc-generation steps are lettered **D1–D5** (pre-flow, not in the 1→41 count). Two parallel tracks — **Analog A1–A9** and **Mixed-signal M1–M4** — run alongside.

Phase → Stage map

- **Phase 1** — Spec & Documents: two input paths (**Agent path** · **doc-gen path D1–D5**)
- **Phase 2** — RTL → Synthesis: Stage 1 (RTL+Verify) · Stage 2 (Synthesis+DFT)
- **Phase 3** — Physical → Tapeout: Stage 3 (Physical+Sign-off) · Stage 4 (Output+Tapeout) · Stage 5 (Manufacturing & Test)
- **Parallel** — Analog A1–A9 · Mixed-signal M1–M4

Phase 1 — Spec & Documents

Two input paths; both yield the same L-series design documents that feed Phase 2:

- `phase1/input_prompt/` — free text / natural language → the **Agent path**
- `phase1/input_doc/` — existing docs / structured YAML → the **doc-gen path** (D1–D5)

Agent path (when the input is free text)

Agent	What it does
PM Agent	Faces the user: turns natural-language requirements into design facts, asks one plain-

Agent	What it does
	language question per gap, confirms, hands off.
IC Expert Agent	Behind the PM Agent: reviews every layer with silicon expertise, fills sensible defaults, cross-checks layer consistency. Never faces the user.

Flow: user free text → PM Agent → IC Expert Agent → finalized L-docs.

doc-gen path D1–D5 (when the input is existing documents)

#	Step	What it does	Tool / How
D1	Ingest & text extraction	Collect the prompt or vendor docs into `input_doc/` and extract plain text.	`phase1_doc_one_shot_runner`
D2	Generate L1–L13 core design-layer docs	Deterministically extract the core design layers (datasheet, spec, regmap, …).	deterministic extractors
D3	Generate L14–L23 docs	Add the protocol / timing / skeleton overlay layers.	overlay extractors
D4	Protocol-class synthesis dispatch	Detect which protocol class (81 classes) the IC belongs to and synthesize its protocol facts.	`is_<proto>` + `<proto>_synth`
D5	Coverage / parity report	Verify the input documents' content fully	`phase1` parity report

#	Step	What it does	Tool / How
		landed in the L-docs.	

Phase 2 — RTL → Synthesis

Stage 1 — RTL Generation & Verification

#	Step	What it does	Tool / How
1	Spec-to-RTL	Author synthesizable RTL from the L-docs.	`spec-to-rtl` skill
2	🔁 Lint	Static-check RTL style and common bugs; auto-fix what can be fixed.	`eda_lint` + `rtl_hygiene_lint`
3	🔁 CDC / RDC check	Verify clock-domain and reset-domain crossings are safe.	`cdc-check` + `rdc-check`
4	🔁 Simulation	Generate testbenches, run functional simulation, measure coverage.	`testbench-gen` + `eda_simulate`
5	🔁 Formal verification	Prove key properties (assertions) always hold.	`formal-verify` + `assertion-gen`
6	FPGA early prototype	Put the design on an FPGA early to verify real behaviour.	`eda_fpga_compile` / `eda_fpga_program` → `.sof`

Stage 2 — Synthesis & DFT

#	Step	What it does	Tool / How
7	Constraint setup	Write timing constraints (SDC) and the PVT corner matrix.	`constraint-gen` → `*.sdc`
8	🔁 SDC validation	Validate the constraints themselves for correctness and completeness.	SDC lint + `sdc_validator_check`
9	Synthesis	Map the RTL to a standard-cell gate netlist.	`eda_synth` + `synth-doctor`
10	🔁 Pre-layout STA	Multi-corner static timing analysis before layout.	`eda_sta` (SS/TT/FF)
11	DFT insertion	Insert scan chains and test logic; generate test patterns (ATPG).	`dft-insert` + `atpg` + `eda_dft`
12	Post-DFT optimization	Re-optimize timing and area after DFT insertion.	resynth / buffering
13	🔁 Equivalence check	Formally prove the gate netlist is functionally equal to the RTL.	`equivalence-check` + Yosys `equiv`

Phase 3 — Physical Design → Tapeout

Stage 3 — Physical Design & Sign-off

#	Step	What it does	Tool / How
14	🔁 pre-PnR Yosys gate	Confirm the synth script and netlist meet PnR requirements before layout.	`yosys_script_template_check` + `yosys_hilomap_required_check`
15	Floorplan + PDN	Plan the chip floorplan and power-delivery network.	`eda_pnr` (init)
16	Clock planning	Plan the clock-tree distribution strategy.	`cts-plan` skill
17	Placement	Place the standard cells (global + detailed).	`eda_pnr`
18	Spare-cell + ECO-prep insertion	Pre-place spare cells and ECO reserves so later bug fixes need only metal-layer changes.	`eda_pnr` + spare-cell checks
19	CTS (Clock Tree Synthesis)	Build the clock tree and balance skew.	`eda_pnr enable_cts=true`
20	🔁 Post-CTS hold fixing	Fix the hold violations the clock tree introduces.	`hold-fix` (`repair_timing -hold`)
21	Routing	Route all signal nets	`eda_pnr enable_detailed_route=true`

#	Step	What it does	Tool / How
		(global + detailed).	
22	Parasitic extraction	Extract post-route parasitic RC (SPEF).	OpenRCX `extract_parasitics`
23	🔁 Post-route STA	Sign-off timing analysis with real parasitics.	`eda_sta` (MMMC)
24	🔁 IR drop	Verify power-grid voltage drop stays within limits.	OpenROAD PSM `analyze_power_grid`
25	🔁 EM check	Check current density to ensure metal-line lifetime.	PSM `-enable_em`
26	🔁 Antenna check	Detect and repair process antenna violations.	OpenROAD `check_antennas` + `repair_antennas`
27	🔁 Signal integrity	Analyse crosstalk / noise impact on signals.	SPEF coupling-cap screen
28	Post-layout gate-level sim	Gate-level simulation with SDF delays to confirm post-layout function.	`eda_simulate`
29	Post-layout SPICE verification	Transistor-level simulation of critical paths	`ams-sim` + `eda_spice`

#	Step	What it does	Tool / How
		and analog blocks.	
30	🔁 Physical verification	DRC / LVS / ERC physical-rule sign-off.	KLayout DRC + LVS + Magic ERC
31	🔁 ECO	Engineering-change repair loop when sign-off finds problems.	`eco-plan`

Stage 4 — Output & Tapeout

#	Step	What it does	Tool / How
32	Power analysis	Analyse full-chip power (pre/post-layout).	`power-analysis`
33	Metal fill	Insert metal fill to meet density rules.	OpenROAD `filler_placement`
34	Tapeout checklist	Walk the final sign-off checklist item by item.	`tapeout-checklist` + `signoff_audit`
35	GDSII output	Produce the GDSII data delivered to the foundry.	`eda_gds` + `def2gds`
36	Foundry handoff	Assemble the complete foundry delivery package.	`tapeout-checklist` + handoff checks
37	FPGA final sign-off	Final FPGA recompile and on-board test.	recompile + on-board test

Stage 5 — Manufacturing & Test (post-fab; fires only when silicon is received)

#	Step	What it does	Tool / How
38	Fabrication	Foundry mask-set and wafer manufacturing (external).	`manufacturing_fab_intake_check`
39	Wafer sort / probe test	Probe-test wafers and pick good dies.	`wafer_sort_yield_check`
40	Packaging	Assembly (wirebond / FC-CSP / WLCSP).	`packaging_intake_check`
41	Final test	Post-package final test (functional + parametric + burn-in).	`final_test_attestation_check`

Parallel tracks

Analog A1–A9 (runs alongside Stages 1–3)

#	Step	What it does	Tool / How
A1	Analog spec extraction	Extract each analog block's spec from the L-docs.	`analog-spec-extract` → `spec.json`
A2	Analog topology selection	Choose a circuit topology per the spec.	`analog-topology-select`
A3	Analog netlist generation	Generate the SPICE netlist.	`eda_xschem_netlist` → `<block>.sp`
A4	Analog corner sweep	Sweep PVT corners to confirm spec	`eda_spice_corner`

#	Step	What it does	Tool / How
		at every corner.	
A5	Analog layout	Complete the analog layout.	`eda_analog_layout`
A6	Analog physical verification	Per-block DRC + LVS before merge.	`analog_a6_block_pv_check`
A7	🔁 Post-layout resimulation	Re-simulate with extracted parasitics; compare pre vs post.	`analog-extraction-resim`
A8	Hardmacro generation	Package as LEF + Liberty + GDS + Verilog and feed Stage 3.	`analog-hardmacro-gen`
A9	🔁 Co-simulation / HW verification	Mixed analog+digital simulation and hardware-in-the-loop.	`mixed-signal-cosim` + `eda_spice`

Mixed-signal M1–M4 (fires when analog blocks are present)

#	Step	What it does	Tool / How
M1	Mixed-signal top-level integration	Merge analog + digital GDS and place macros.	`mixed_signal_merge_check`
M2	Mixed-signal power domain verification	Check level-shifters / isolation across power domains.	power-domain checks
M3		AMS co-simulation	`mixed_signal_cosim_check`

#	Step	What it does	Tool / How
	Mixed-signal verification	and interface signal integrity.	
M4	Mixed-signal sign-off	Top-level physical verification and final verdict.	`mixed_signal_signoff_check`

Totals

Phase	Stages	Steps
Phase 1 — Spec & Documents	two input paths (Agent · doc-gen)	D1–D5 + PM Agent · IC Expert Agent
Phase 2 — RTL → Synthesis	Stage 1 · Stage 2	1–13
Phase 3 — Physical → Tapeout	Stage 3 · Stage 4 · Stage 5	14–41
Parallel	Analog · Mixed-signal	A1–A9 · M1–M4

41 sequential steps (Stage 1: 1–6 · Stage 2: 7–13 · Stage 3: 14–31 · Stage 4: 32–37 · Stage 5: 38–41), plus Phase 1 (Agent path and doc-gen path D1–D5) and the two parallel tracks (Analog A1–A9 · Mixed-signal M1–M4). Pre-flight: P0 (environment health check). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

繁中版：`ALL_STEPS_v2.2.0.zh-TW.md`.